

PAPER_1121: Interstellar Shock-Driven Prestellar Core Collapse and Prebiotic Molecule Release

Abstract

Following Ceccarelli & Codella (2024), we implement a UQFF calculator for interstellar shock-triggered prestellar core collapse and molecule release. J-type shocks produce abrupt density compressions $S(t)$ with high sputtering efficiency ($\eta \sim 0.9$), while C-type shocks enable gradual molecule release $C(t)$ for SiO, H₂O, and formamide. Prestellar core conditions ($T \sim 10\text{-}20\text{ K}$, $n \sim 10^5\text{-}10^6\text{ cm}^{-3}$) are modeled as [SCm]-[UA] interactions within the UQFF g_{Shock} framework.

1. Introduction

Interstellar shocks play a fundamental role in star formation by compressing molecular gas past the Jeans threshold. The UQFF models this via:

$$g_{\text{Shock}} = \frac{GM}{r^2} \cdot S(t) + C(t)$$

where $S(t)$ is the compression factor and $C(t)$ represents molecule release from grain mantles.

2. Shock Classification

J-type Shocks

- Abrupt velocity discontinuity
- Mach number $\mathcal{M} = v_s/c_s$, compression $S(t) = 4\mathcal{M}^2$
- High sputtering: $\eta_{\text{sput}} \sim 0.9$
- SiO release from grain core destruction

C-type Shocks

- Continuous, magnetically mediated compression
- Typical $S(t) \sim 10$
- Lower sputtering: $\eta_{\text{sput}} \sim 0.1$
- H₂O and formamide release from ice mantles

3. Prebiotic Molecule Release

Post-shock abundances:

$$X(\text{SiO})_{\text{post}} = X_0 + \eta_{\text{sput}} \times 10^{-8}$$

$$X(\text{H}_2\text{O})_{\text{post}} = X_0 \cdot (1 + \eta \cdot S(t) \cdot 0.01)$$

$$X(\text{formamide})_{\text{post}} = X_0 \cdot (1 + \eta \cdot S(t) \cdot 0.5)$$

4. Jeans Mass Evolution

Post-shock Jeans mass determines whether collapse proceeds:

$$M_J = \left(\frac{\pi c_s^2}{G} \right)^{3/2} \rho_{\text{post}}^{-1/2}$$

5. UQFF Alignment

Overall alignment: **80%**. The [SCm]-[UA] interaction framework successfully models both shock-induced compression and the prebiotic chemistry pathway critical for origins of life.

References

- Ceccarelli, C. & Codella, C. (2024). Interstellar Shocks and Star Formation Chemistry.
- arXiv:2404.19533 — J-type Shocks in Perseus Molecular Cloud.